Team Details

Team Name: **MAAVForge**

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	Team Leader	**Maadeshwar DV**	*3rd year*
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3	Member 2	**Anirudh JP**	*3rd year*
4	Member 3	**Vanshita Patil**	*3rd year*

COLLEGE NAME

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Problem Statement Addressed

AI-Based Restoration of Degraded Images for Semiconductor Inspection

DESCRIPTION / DETAILS

Semiconductor inspection images contain microscopic structures where even small amounts of noise or lost resolution can hide defects. MAAVForge restores degraded grayscale images by jointly suppressing speckle and Gaussian-like noise while reconstructing 2× spatial resolution.

The objective is to recover sharp, inspection-relevant detail from a degraded 128×128 input and reconstruct a clean 256×256 output, while remaining robust to unseen image sources.

Idea Description

KEY CONCEPT & APPROACH

Frequency-Aware NAFNet-SR: a single end-to-end restoration network that learns denoising and 2× super-resolution together. NAFNet is lightweight and efficient, while frequency-aware learning helps preserve genuine microscopic edges instead of simply smoothing noise.

SOLUTION OVERVIEW

Pipeline: Noisy 128×128 input → NAFNet encoder–decoder → PixelShuffle ×2 + bicubic residual → restored 256×256 output.
It addresses speckle, Gaussian-like degradation and resolution loss simultaneously. At inference, 8-fold geometric test-time augmentation improves robustness by averaging inverse-transformed predictions.

Proposed Solution

SOLUTION DETAILS

Architecture: Conv (1→32) → encoder with 2/2/4 NAFBlocks → 6-block bottleneck → decoder with 4/2/2 NAFBlocks → PixelShuffle ×2 + bicubic residual.

Training: 2,880 paired images, 320 validation images, AdamW, cosine annealing, AMP, gradient clipping and early stopping. Paired augmentation uses horizontal/vertical flips and 90° rotations.

Composite loss: L = 1.0·Charbonnier + 0.1·SSIM + 0.05·Focal Frequency. The three terms jointly optimize pixel accuracy, structural similarity and frequency-domain detail.

Innovation and Uniqueness

KEY INNOVATION

Frequency-aware restoration. Focal Frequency Loss explicitly matches the restored and ground-truth frequency content, helping suppress broadband noise while retaining meaningful high-frequency semiconductor structures.

The model also uses a robust Charbonnier + SSIM + FFL composite objective and 8-fold geometric TTA.

COMPETITIVE ADVANTAGE

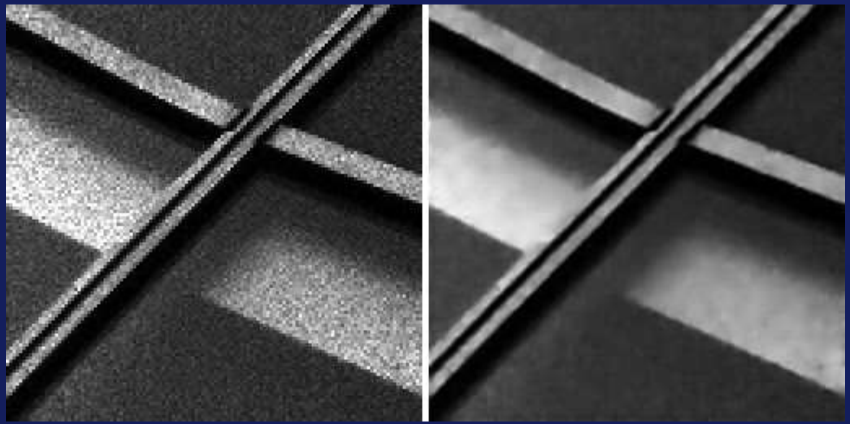
Unlike a slow two-stage “denoise then upscale” pipeline, MAAVForge performs both tasks in one compact model.

- ~4.26M parameters
- PixelShuffle avoids transposed-convolution artifacts
- SimpleGate-based NAFNet blocks are efficient
- ~62 ms/image on NVIDIA T4
- Benchmark-friendly standalone inference pipeline

Results

Validation performance and direct visual evidence from MAAVForge restoration outputs.

Visual Evidence (Test Set Sample: 000041.npy)"
Left: Original degraded 128×128 input (Multiplicative Speckle Noise). Right: MAAVForge 256×256 reconstruction (Denoised & 2x Super-Resolution).



Validation Metrics

PSNR ↑ 26.97 dB

SSIM ↑ 0.7570
LPIPS: not evaluated

~62 ms/image on NVIDIA T4
100 epochs • ~1 h 45 min training

Technology & Feasibility/Methodology Used

IMPLEMENTATION STRATEGY

Software: Python 3.10+, PyTorch 2.0+, CUDA, Automatic Mixed Precision.

Model: NAFNet-SR, 4,257,412 parameters, grayscale 128×128 → 256×256.

Training: NVIDIA T4 GPU, batch size 8, 100 epochs, AdamW + cosine annealing. Standalone evaluation script accepts an input directory and output directory with no manual edits.

Feasibility: compact architecture and reproducible scripts are designed for fast benchmarking and deployment.

NAFNet-SR
PyTorch + CUDA

NVIDIA T4
GPU Training

Python
GitHub

GitHub & Video Link

GitHub Repository

https://github.com/Maadeshwar/KLA-FARR-Restoration

Prototype / Simulation Video

https://drive.google.com/file/d/1lM5HDFQPe_RUyjsTQRMbDymVJWtwDH-g/view

Research and References

Research Background & Methodology

MAAVForge is grounded in efficient convolutional image restoration, robust pixel-domain reconstruction, structural similarity optimization and frequency-domain loss design. The architecture is tailored to simultaneous noise suppression and 2× super-resolution of paired grayscale semiconductor images.

References & Citations

Chen et al. — Simple Baselines for Image Restoration (NAFNet), ECCV 2022.

Shi et al. — Real-Time Single Image and Video Super-Resolution Using Efficient Sub-Pixel Convolution.

Wang et al. — Image Quality Assessment: From Error Visibility to Structural Similarity (SSIM).